

Technics
Panasonic
Quasar®

Technical Guide

Purposes of CD player Adjustments (One-Beam System)

Audio Division
Sales Engineering Dept.
Matsushita Electric Industrial Co., Ltd.

CONTENTS

I. MECHANICAL ADJUSTMENTS 1~4

- (1) Turntable Height adjustment 1~2
- (2) Mechanical adjustment 3~4

II. ELECTRICAL ADJUSTMENTS 5~15

- (1) Best eye (PD balance) adjustment 5~6
- (2) Focus gain adjustment 7~8
- (3) Tracking gain adjustment 9
- (4) Focus offset adjustment 10~11
- (5) Tracking offset adjustment 12
- (6) Tracking offset balance adjustment 13~14
- (7) PLL adjustment 15

III. VARIOUS DEFECTS OF DISC & RF WAVEFORM .. 16

C. Basic principle of height adjustment of turntable

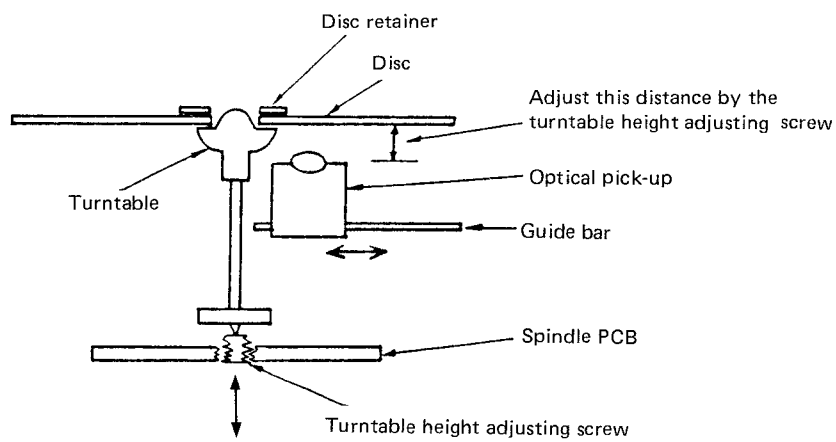


Fig. 1-1-2 DD motor

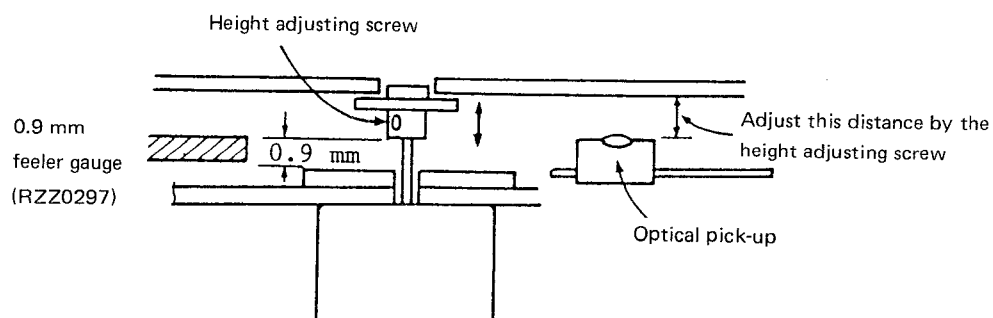


Fig. 1-1-3 DC Motor

2. Mechanical adjustment (optical pick-up)

A. Purpose

Unless the lens is perpendicular to the disc surface, the laser beam may not be focused. The mechanical alignment is therefore necessary to adjust the optical pick-up lens so that the lens is at an optimum perpendicular position.

B. Basic principle of mechanical adjustment

To accomplish the above condition, it is necessary to adjust Hex screws (A) and (B) as shown.

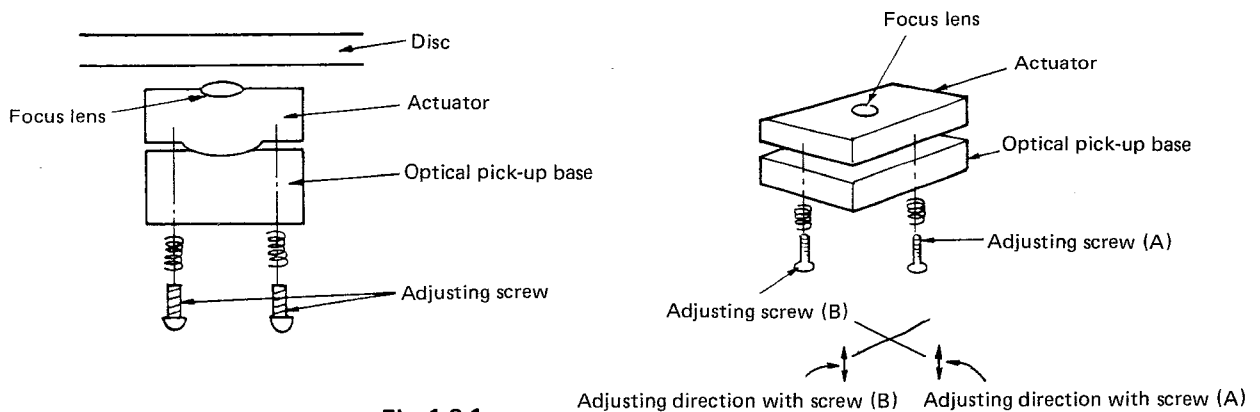


Fig. 1-2-1

C. Basic principle of adjustment method

The warped Test Disc (SZZP1056C) is used to simulate the non-perpendicular condition. While playing this Disc, the RF amplitude's vertical fluctuation is to be minimized; at which point the most perpendicular point is set.

1. RF waveform at use of test disc (SZZP1056C)

A thin piece of metal is mounted to the part where the disc touches the turntable. The RF waveform is fluctuated vertically (amplitude fluctuated)

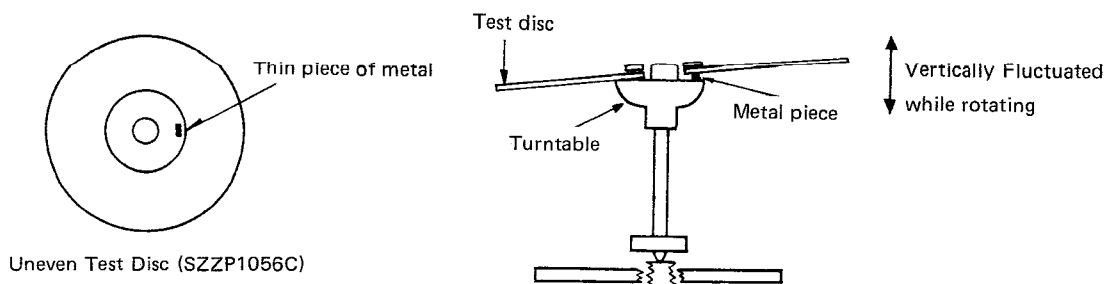


Fig. 1-2-2

2. RF waveform when properly aligned

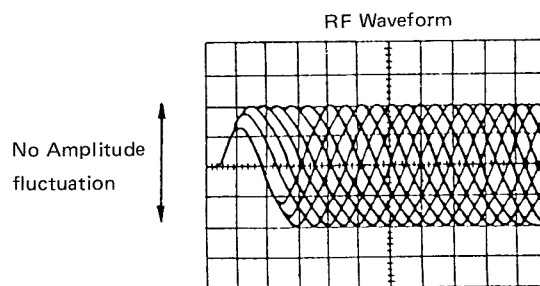
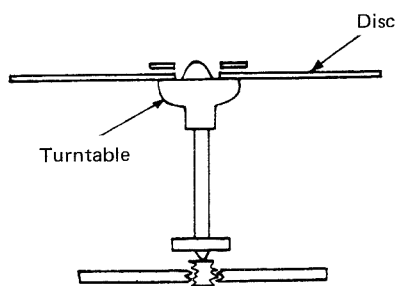
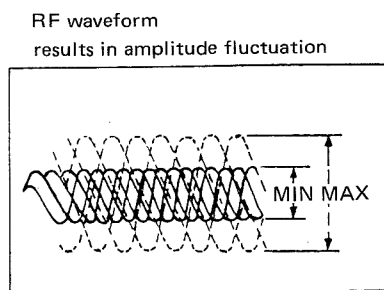
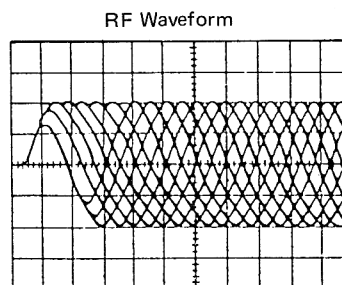
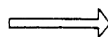


Fig. 1-2-3

RF waveform is stable,
free of vertical amplitude fluctuation.



Waveform before alignment



Waveform of optimum alignment

Fig. 1-2-4

II. ELECTRICAL ADJUSTMENT

These adjustments are so important to the extent that the performance of the CD player depends mainly on their accuracy.

1 Best eye (PD balance) adjustment

A. Purpose

Due to fluctuations of as described earlier, under "turntable height adjustment" the focus servo is designed to work within a certain operating range. The purpose of this adjustment is to set the focus servo's reference voltage (zero); at which point the lens focusing depth is set.

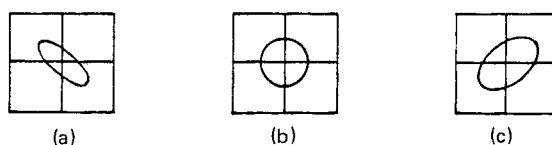


Fig. 2-1-1 Beam shape received by the photodetector

B. Basic principle of best eye adjustment & Best point

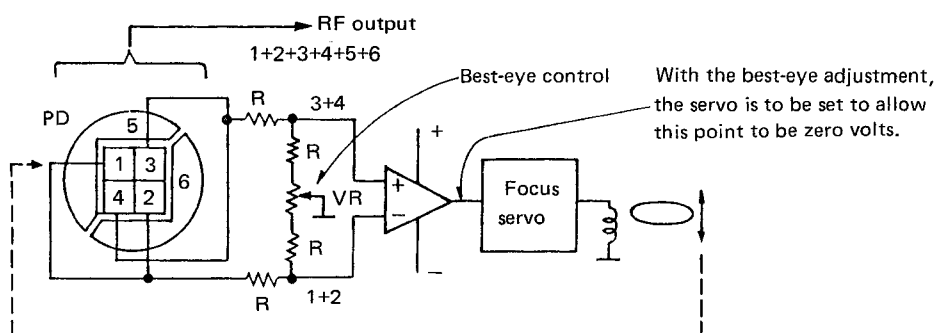
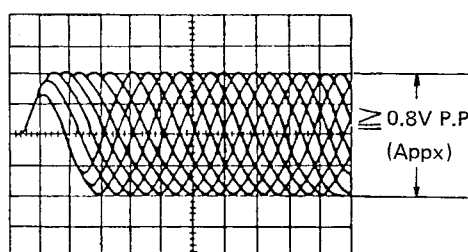


Fig. 2-1-2 Focus servo loop



Maximum amplitude of the RF waveform is achieved when the best eye adjustment is properly set, as described above.

Fig. 2-1-3 RF Waveform

C. Basic principle of adjustment

Due to distance variations between the disc surface and the lens caused by uneven discs, the laser beam covering a given pit varies in diameter. When the lens is close to the Disc the diameter is large and covers adjacent pits. The signal from adjacent pits causes interference and the RF becomes distorted.

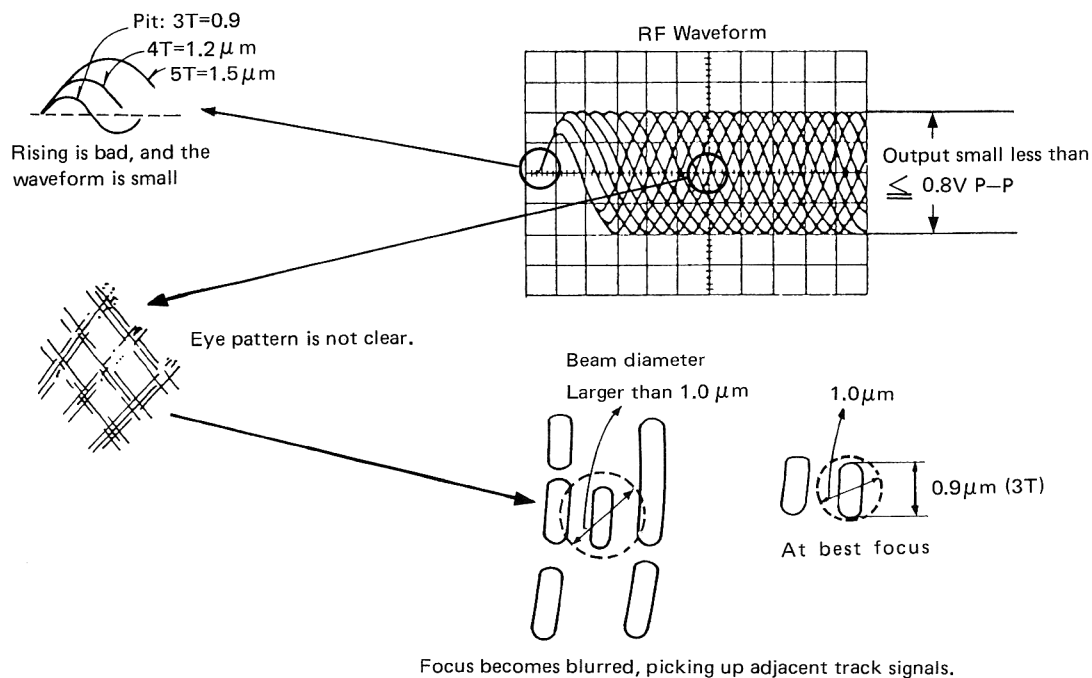


Fig. 2-1-4

D. Adjustment method

It is required to adjust so that the RF output is maximum and the eye pattern appears clearly.

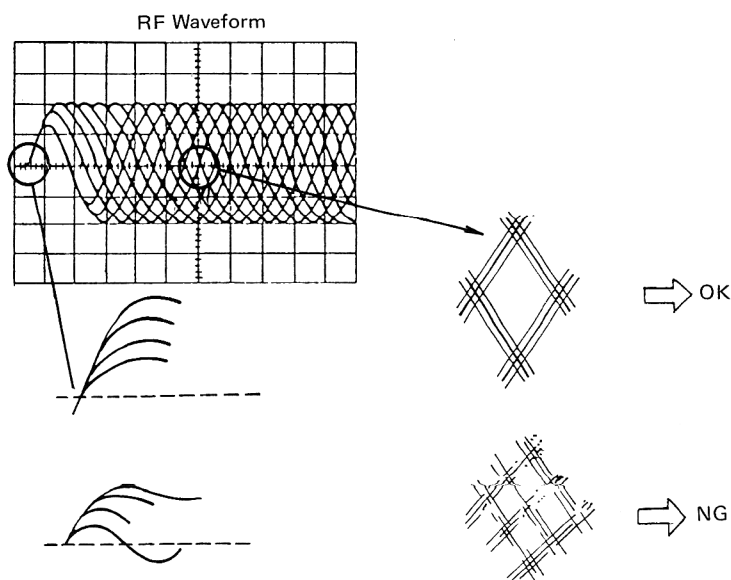


Fig. 2-1-5

2. Focus gain adjustment

A. Purpose

Regardless of disc warpage ($\pm 0.4\text{mm}$) due to manufacturing tolerances, the focus servo must operate to constantly keep the lens focused on the disc's pits. The greater the servo's gain, the easier the lens can focus on the disc; however it becomes unstable due to unwanted signals. The gain is therefore set to unity ($0\text{ dB}=1$, @750Hz) at which point stability is acceptable and the gain is adequate.

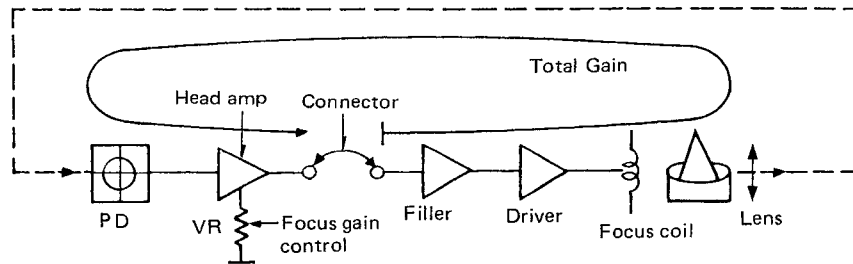


Fig. 2-2-1 Focus Servo Loop

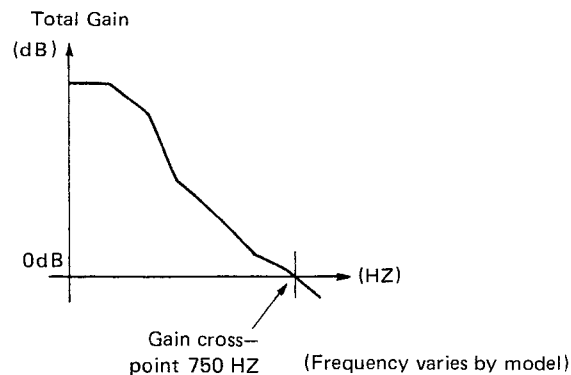


Fig. 2-2-2 Focus servo gain characteristics

B. Principle of the focus gain adjustment

The focus servo loop is opened, at which point a 750 Hz signal is applied to simulate the acceptable condition. This point is TP2 on the "servo gain adjuster". The signal, once through the focus servo, appears at TP1. The focus gain variable resistor is used to set the voltage at TP1 to equal the level at TP2; at which point the gain is one (0 dB). The reason why the focus servo must be adjusted using a 750 Hz tone is because this is the frequency that causes the gain to become unity (0dB=1).

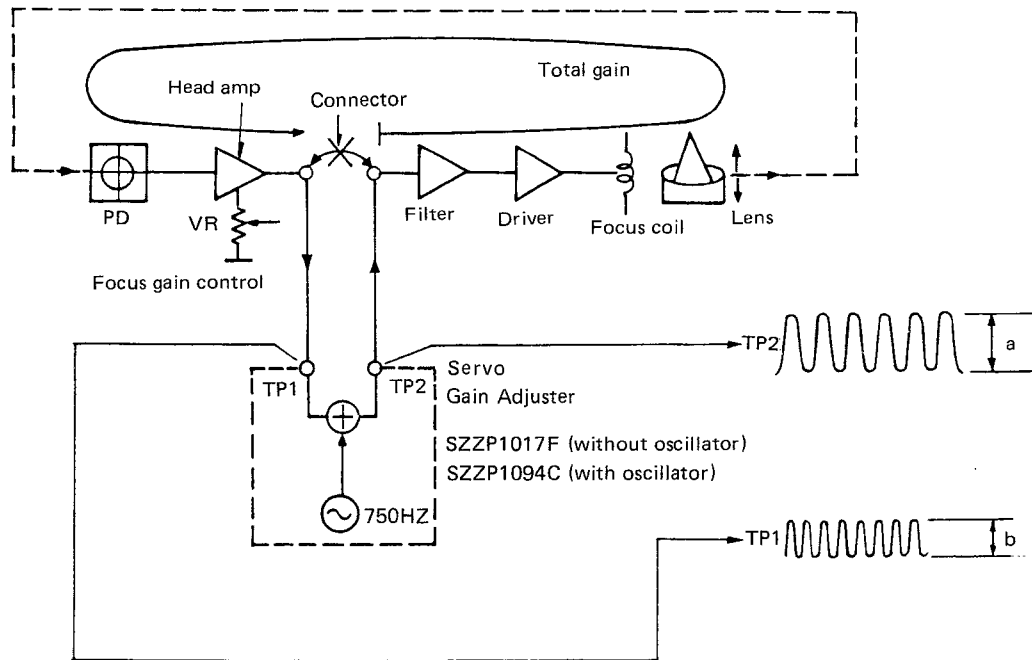


Fig. 2-2-3

Note: Explanation was made based on 750Hz gain cross-point frequency.
This frequency may differ depend on model.
Please refer to the service manual of the corresponding model for frequency.

3. Tracking gain adjustment

A. Purpose

During disc manufacturing, disc eccentricity results due to tooling tolerances. Eccentricity also results if the disc does not rest exactly in the center of the CD player's spindle.

The tracking servo is designed to overcome these tolerances and keep the lens on track.

B. Principle of the Tracking gain adjustment

The tracking servo loop is opened at which point a 1KHz, 1.2KHz or 1.5KHz signal is fed. This point is TP2 of the "servo gain adjuster".

The signal, once through the servo, appears at TP1. The tracking gain control is to be adjusted so that the level of TP1 equals that of TP2, at which point the gain is one (0dB)

The reason why the focus servo to be adjusted using a 1KHz, 1.2KHz or 1.5KHz tone is because these are the frequencies that cause the gain to become unity (0dB=1).

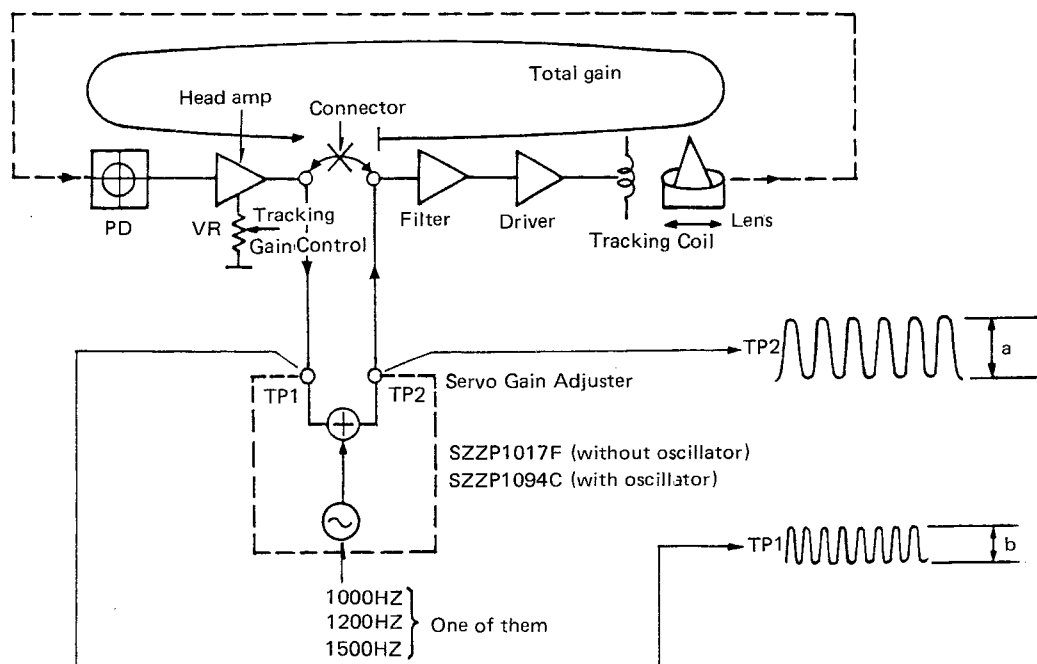


Fig. 2-3-1

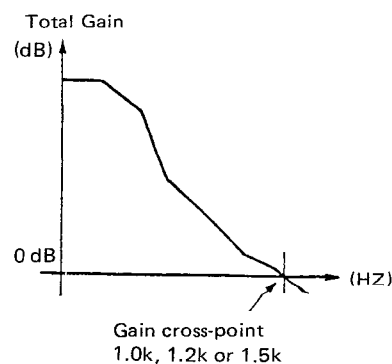


Fig. 2-3-2 Tracking servo characteristics

4. Focus offset adjustment

A. Purpose

The CD player must be able to compensate for manufacturing imperfections of disc and dust or any other contaminants on the signal side.

As the pick-up traces the track, if dust or disc imperfections are present, the focusing depth changes.

Without proper focus the tracking servo is affected as well; causing the pick-up to move to a different track.

When the lens detects imperfections as mentions above, the focus servo is enabled in order to maintain proper focusing within the operating range of the servo.

The focus offset alignment is therefore necessary to adjust the servo's operating reference point.

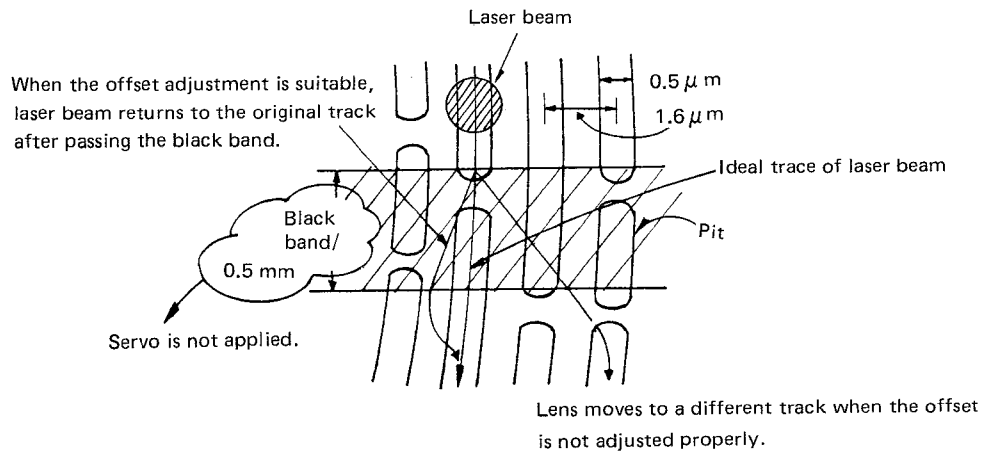


Fig. 2-4-1

B. Test disc, RF waveform and FE signal.

A black band disc is used to simulate disc imperfection or dust particles, causing the RF to appear as in fig. 2-4-4 (a) and (b).

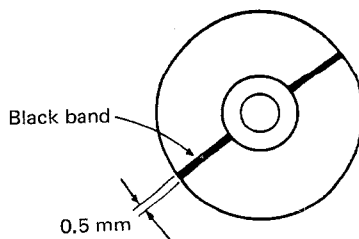


Fig. 2-4-2
Test disc (SZZP1057C)

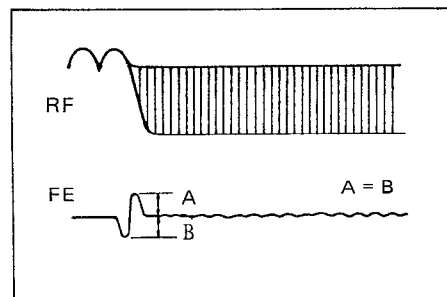


Fig. 2-4-3
RF Waveform when properly adjusted

C. Focus offset adjustment method

It should be adjusted so that the RF waveform dip is minimized or eliminated and the size of FE ① and ② is at minimum or balanced.

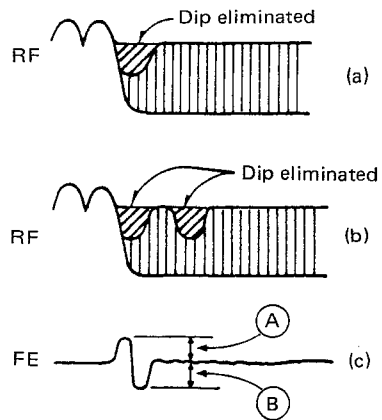


Fig. 2-4-4

5. Tracking offset adjustment

A. Purpose

As with the focus offset adjustment, the tracking offset adjustment is necessary to set the reference point of the tracking servo operating range.

The only difference is that the tracking servo compensate for horizontal control as opposed to vertical control by the focus servo,

B. Test disc, RF waveform and TE (Tracking Error) signal

The test disc, RF waveform, and TE signal waveform are the same as described in the focus offset adjustment. The only difference is the TE signal test point.

C. Tracking offset adjustment method

The RF waveform must be adjusted such that the dip(s) is (are) minimized or eliminated, as indicated below in figures (a) and (b). The TE waveform must be adjusted such that (A) and (B) are equal, as indicated in figure (c) below.

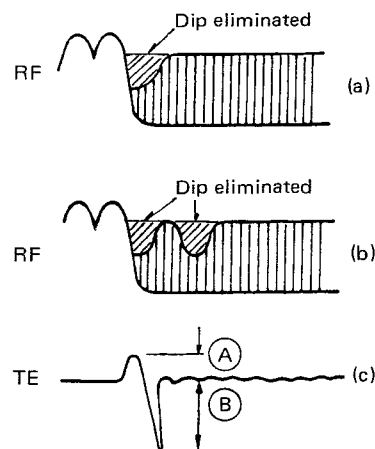


Fig. 2-5-1

6. Tracking offset balance adjustment

A. Purpose

During playback operation, the laser beam must follow the track exactly at its center. If it is not, proper tracking can not be accomplished, and skipping may occur during playback of the disc which has manufacturing imperfections and dust or any other contaminants on the signal side.

B. Basic principle of adjustment

A 1 KHz signal is applied to the tracking servo circuit to intentionally cause the beam to oscillate from side to side of the track being traced (see fig. 2-6-1). This is done to simulate an error signal so that the tracking servo can be adjusted for minimum gain (minimum swing) at 1 KHz.

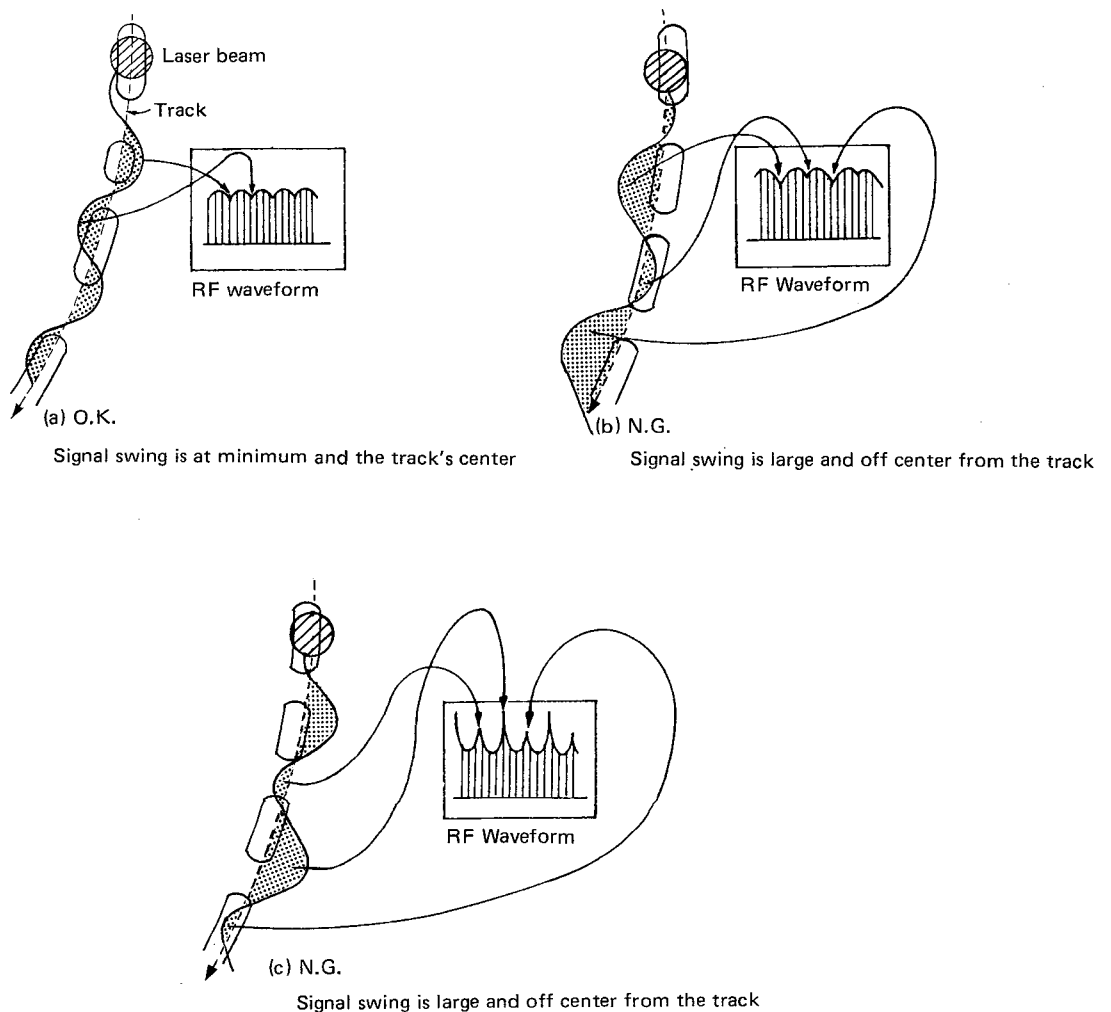


Fig. 2-6-1 Beam movement with error signal

C. Adjustment method

The 1 KHz signal is fed through the tracking servo. The tracking offset balance control is to be adjusted for minimum and balanced level on the RF waveform.

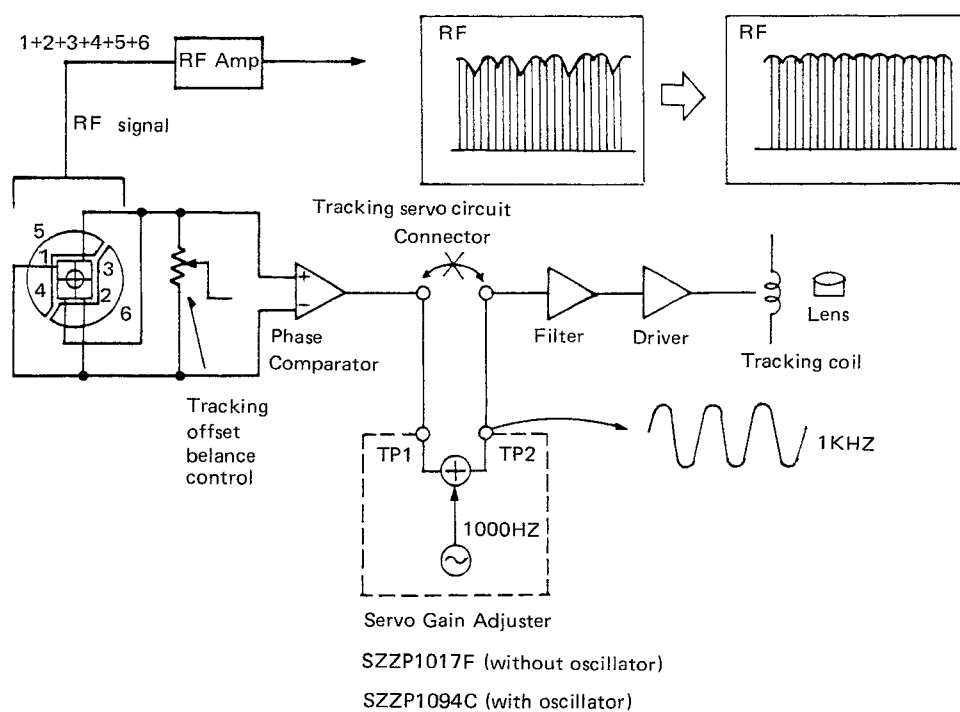


Fig. 2-6-2

7. PLL adjustment

A. Purpose

The purpose of the PLL adjustment is to set the VCO (Voltage controlled Oscillator) circuit such that the 4.3218 MHz clock will lock within its operating range.

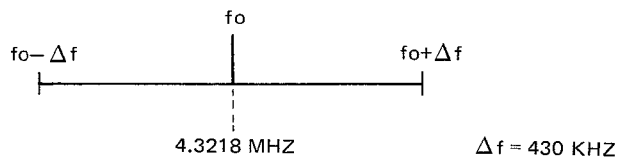
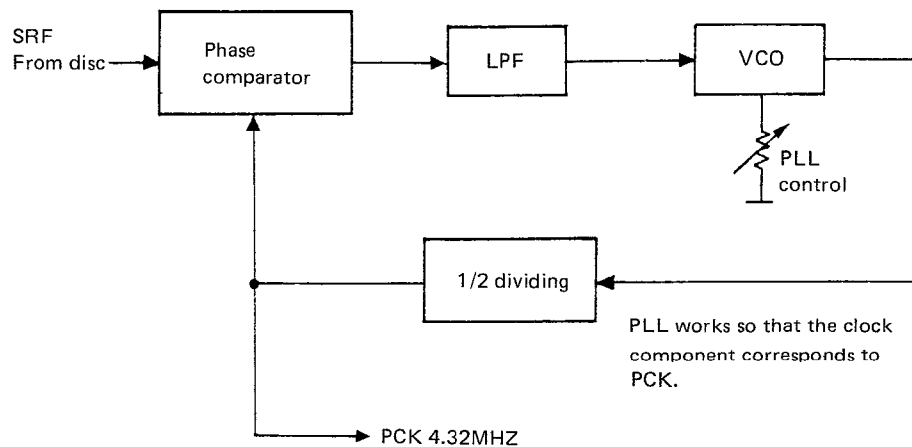


Fig. 2-7-1 PLL operating range

The PLL (Phase Locked Loop) reference frequency is constantly compared to the disc's sync data (extracted from SRF) representing a frequency of 4.3218 MHz. When these two frequencies are equal the PLL is locked. If the PLL is not adjusted within its operating range it can cause incorrect disc rotation or dropout.



III. VARIOUS DEFECTS OF DISC & RF WAVEFORM

Defects related to the disc are shown in Fig. (A). Their respective RF waveforms are shown in Fig. (B).

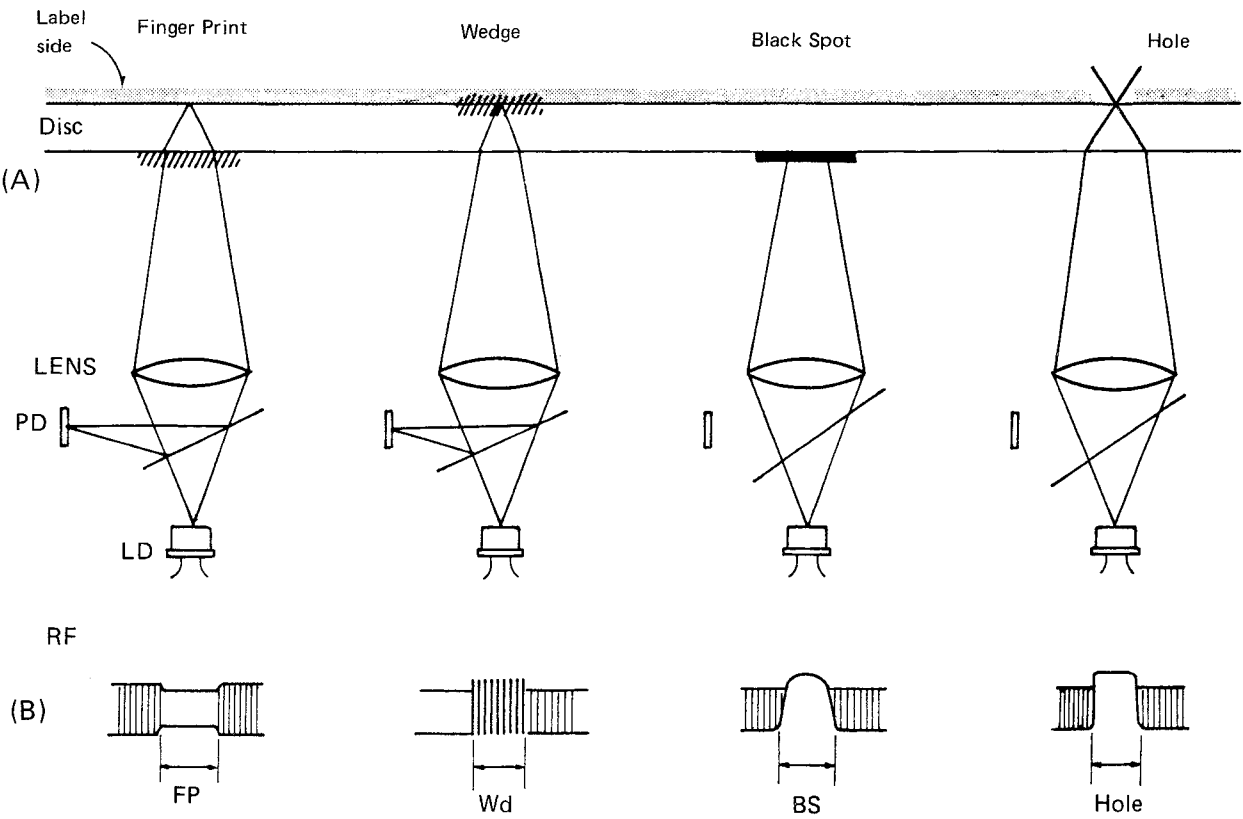


Fig. 3-1-1

